

Dr. Charles E. Mentzer
Cmentzer@gapp.nthu.edu.tw

Education:

University of Missouri – Columbia August 2019-July 2025
DEGREE: Master of Science Physics (May 2022) (3.929)
DEGREE: Ph.D. Physics
CONCENTRATION: Astrophysics

University of South Florida August 2014 -May 2018
DEGREE: Bachelor of Science
CONCENTRATION: Physics
MINORS: Mathematics & Astronomy

Experience:

NTHU Postdoctoral Research Fellow/ National Tsing Hua University/ August 2025-Present
Investigations of PAHs, ices, and dust and their roles in the physics of planet-forming disks.

Adjunct Faculty/ Moberly Area Community College/ August 2022- January 2025
Teaching PHY101 Introduction to Physical Science, PHY162 General Astronomy. In-person Physics Lab and curriculum development for both classes.

Outreach Coordinator/University of Missouri LAWS Observatory/Aug 2021- July 2025
General maintenance of equipment and orchestration of community and outreach events through the LAWS Observatory.

Research Assistant / Dust and PAH Group under Dr. Aigen Li / May 2020 – July 2025
Theoretical studies of the interstellar medium (with a focus on interstellar dust and polycyclic aromatic hydrocarbons (PAHs)), the formation of planetary systems (with a focus on protoplanetary dust disks around young stars), vibrational excitation of PAHs, and investigation of merging galaxies involving AGN, starburst, and Quasar host galaxies through JWST NIRSpec and MIRI/MRS.

Teaching Assistant/ University of Missouri Physics Department /August 2019 – May 2025
Running labs and problem-solving discussions for Physics 1 & 2 with and without calculus. Running and developing ASTRON 1020 - Introduction to Laboratory Astronomy an introductory course that surveys astronomical methods, instruments, observations, and measurement techniques.

Research Assistant / Computational Condensed Matter Group under Dr. Inna Ponomareva / August 2018 – May 2019
Conducting Molecular Dynamic Simulations of Ferroelectric Relaxor $\text{Ba}(\text{Ti}_{1-x}\text{Zr}_x)\text{O}_3$ using supercomputer facilities
[<http://shell.cas.usf.edu/~iponomar/index.html>]

Curriculum Development / Laboratory development with Dr. Alexander McCormick / December 2017 – May 2018
Utilized circuitry and radio astronomy to detect radiation from the Sun to create and implement an exercise for upper-level Astronomy students

Research Assistant / Planetary Science Group under Dr. Maria Womack / June 2017 – May 2018
Research into the visual brightness behavior of Comet C/1995 O1 (Hale-Bopp), including analyzing its calibrations and statistics, calculating dust production and total CO production models, then submitting to NASA archive.
[<https://www.mariawomack.com/>]

Publications:

JWST Edge-on Disk Ice (JEDIce): An Analysis of PAH Signatures in Edge-on Class II T Tauri Disks (In Prep)

Charles Mentzer, Daniel Harsono, Jennifer Bergner, Jennifer Noble, Emmanuel Dartois, Zhi-Yun Li, Korash Assani, Lukas Welzel, Mayank Narang, Julia Santos, Nicole Arulanantham, Klaus Pontoppidan, Maria Drozdovskaya, Melissa McClure, Karin Öberg, Yao-Lun Yang, Will Thompson, Elizabeth Yunerman, Aditya Arabhavi, Alice Booth

Widespread Detection of Aromatic and Aliphatic Emission in the Dual Quasar J0749+2255 at Cosmic Noon (July 2026)

C. E. Mentzer, Aigen Li, X. J. Yang

Publication: The Astrophysical Journal Letters, Volume 1005, Issue 2, id.L38, 10 pp.

[<https://arxiv.org/abs/2607.03913>]

Phase Evolution in Ba(Ti_{1-x}Zr_x)O₃ Ferroelectric Relaxor from Atomistic Simulations (25 February 2019)

C. Mentzer, S. Lisenkov, Z. G. Fthenakis, and I. Ponomareva.

Publication: Phys. Rev. B 99, 064111

[<https://journals.aps.org/prb/abstract/10.1103/PhysRevB.99.064111>]

Advanced Manuscripts Awaiting Submission:

On the Aliphatic Contents of PAHs in VV 114 as Revealed by JWST/NIRSpec

Charles Mentzer, Aigen Li

Modelling the PAH Emission in VV 114

Charles Mentzer, Aigen Li

Red Stars over PAHs: Excitation of PAH Emission in UV-Poor Environments

Charles Mentzer, Aigen Li

Spatially Resolved PAH Emission in the Protoplanetary Disk around HD 97048

Charles Mentzer, Aigen Li, Gidi Yoffe, Roy Van Boekel, Thomas Henning, Koen Maaskant, L.B.F.M. Waters

Spatially Resolved PAH Emission in the Protoplanetary Disk around HD 169142

Charles Mentzer, Aigen Li

PAH Emission in the Circumstellar Envelopes around R Coronae Borealis Stars

K.J. Li, C. Mentzer, A. Li

JWST Edge-on Disk Ice (JEDIce): Vibrationally hot, rotationally cold H_2 in the outer disk of Oph 163131 non-thermally excited by UV and cosmic rays (July 2026)

Assani, Korash ; Li, Zhi-Yun ; Bergner, Jennifer B. ; Neufeld, David A. ; Harsono, Daniel ; Drozdovskaya, Maria N. ; Padovani, Marco ; Dartois, Emmanuel ; Noble, Jennifer A. ; Arulanantham, Nicole ; Booth, Alice S. ; Yang, Yao-Lun ; Narang, Mayank ; Thompson, Will E. ; Yunerman, Elizabeth ; Öberg, Karin I. ; Santos, Julia C. ; Mentzer, Charles ; Ramsey, Jon P. ; Welzel, Lukas ; Pontoppidan, Klaus M. ; McClure, Melissa

Publication:

[<https://arxiv.org/abs/2607.09407>]

JWST Edge-on Disk Ice (JEDIce): Program overview and ice survey results (March 2026)

Jennifer B. Bergner, Nicole Arulanantham, Emmanuel Dartois, Maria N. Drozdovskaya, Daniel Harsono, Melissa McClure, Jennifer A. Noble, Karin I. Öberg, Klaus M. Pontoppidan, Yao-Lun Yang, Korash A)ssani, Zhi-Yun Li, Julia C. Santos, Will E. Thompson, Lukas Welzel, Elizabeth S. Yunerman, Aditya M. Arabhavi, Alice S. Booth, Charles Mentzer, Mayank Narang, Thomas Henning, Inga Kamp, Giulia Perotti, Alice Somigliana

Publication: The Astrophysical Journal, Volume 1002, Issue 1, id.17, 18 pp.

[<https://arxiv.org/abs/2603.18163>]

The Visual Lightcurve of Comet C/1995 O1 (Hale-Bopp) from 1995 to 1999 (February 2021)

Womack, M ; Curtis, O. ; Rabson, D. A. ; Harrington Pinto, O.; Wierzechos, K.; Cruz Gonzalez, S.; Sarid, G.
Mentzer, C. ; Lastra, N. ; Pichette, N.; Ruffini, N. ; Cox, T. ; Rivera, I. ; Micciche, A. ; Jackson, C. ; Homich, A. ;
Tollison, A. ; Reed, S. ; Zilka, J. ; Henning, B. ; Spinar, M. ; Escoto, S. Rosslyn ; Erdahl, T. ; Goldschen-Ohm,
Marcel P. ; Uhl, W. T.

Publication: The Planetary Science Journal, Volume 2, Issue 1, id.17, 16 pp.

[<https://ui.adsabs.harvard.edu/abs/2021PSJ.....2...17W/abstract>]

Conferences:

Contributed Oral Presentation at Exploring the aromatic universe in the JWST era London, Canada July 2026

JWST Edge-on Disk Ice (JEDIce): An analysis of PAH signatures within the disks

Charles Mentzer, Daniel Harsono, Jennifer Bergner, Jennifer Noble, Emmanuel Dartois, Zhi-Yun Li, Korash Assani,
Lukas Welzel, Mayank Narang, Julia Santos, Nicole Arulanantham, Klaus Pontoppidan, Maria Drozdovskaya,
Melissa McClure, Karin Öberg, Yao-Lun Yang, Will Thompson, Elizabeth Yunerman, Aditya Arabhavi, Alice
Booth

Contributed Oral Presentation at The Astronomical Society of the Republic of China (Taiwan) May 2026

JWST Edge-on Disk Ice (JEDIce): An analysis of PAH signatures within the disks

Charles Mentzer, Daniel Harsono, Jennifer Bergner, Jennifer Noble, Emmanuel Dartois, Zhi-Yun Li, Korash Assani,
Lukas Welzel, Mayank Narang, Julia Santos, Nicole Arulanantham, Klaus Pontoppidan, Maria Drozdovskaya,
Melissa McClure, Karin Öberg, Yao-Lun Yang, Will Thompson, Elizabeth Yunerman, Aditya Arabhavi, Alice
Booth

Contributed Oral Presentation at Annual Meeting of the Physical Society of Taiwan January 2026

Tracing the Missing Sulfur and Deuterium: The Role of Modified PAHs in the Interstellar Medium

Charles Mentzer, Aigen Li, X. J. Yang

Contributed Oral Presentation at Laboratory Astrophysics Division special session “Contributions to JWST Studies of Star Formation and the Interstellar Medium” at American Astronomical Society January 2025 National Harbor, Maryland:

Aromatics in Merging Galaxies Involving Active Galactic Nuclei and Starburst Cores as Probed by the James Webb Space Telescope

Charles Mentzer, Aigen Li

Contributed Oral Presentation at The 14th meeting on Cosmic Dust Aug 2024 Nagoya, Japan:

Aromatics in Merging Galaxies Involving Active Galactic Nuclei and Starburst Cores as Probed by the James Webb Space Telescope

Charles Mentzer, Aigen Li

[https://www.cps-jp.org/~dust/Program_XIV_files/dust14-d4_02.pdf]

Contributed Oral Presentation at The 13th meeting on Cosmic Dust Aug 2023 Kitakyushu, Japan:

Red Stars over PAHs

Charles Mentzer, Aigen Li

[https://www.cps-jp.org/~dust/Program_XIII_files/dust13-d2-07.pdf]

Infrared Emission Spectra of R Coronae Borealis Stars

Aigen Li, J.R. Lin, K.J. Li, C.E. Mentzer, W.Y. Cui, J.R. Shi

[https://www.cps-jp.org/~dust/Program_XIII_files/dust13-d2-06.pdf]

Contributed Oral Presentation at American Astronomical Society June 2022 Pasadena, California:

Spatially Resolved PAH Emission in the Protoplanetary Disk around HD 97048

Charles Mentzer, Aigen Li, Gidi Yoffe, Roy Van Boekel, Thomas Henning, Koen Maaskant, L.B.F.M. Waters
[<https://submissions.mirasmart.com/AAS240/Itinerary/PresentationDetail.aspx?evdid=11>]

Contributed Oral Presentation at Fundamental Physics of Ferroelectrics and Related Materials 2019 Tampa, Florida:

Phase Evolution in Ba(Ti_{1-x}Zr_x)O₃ Ferroelectric Relaxor from Atomistic Simulations

C. Mentzer, S. Lisenkov, Z. G. Fthenakis, and I. Ponomareva.

[<https://www.materialsbydesign.org/schedule2019>]

Interactive Display at American Astronomical Society October Meeting 2017 Provo, Utah:

Hale-Bopp Visual Lightcurve

Womack, Maria; Curtis, A.; Lastra, N.; Harrington Pinto, O.; Rabson, D. A.; Wierzechos, K.; Cox, T.; Rivera, I.; Mentzer, C.; Ruffini, N.; Jackson, C.; Micciche, A. NASA Planetary Data System, PDS4 Data Collection, 2018:

[<https://tinyurl.com/y9qfh15a>]

Proposals:

JWST Cycle 5 Proposals (Unsuccessful):

Principal Investigator:

Proposal 10949: PAH Emission in the High-Latitude Translucent Cloud towards HD 210121: A sightline with Exceedingly Anomalous Ultraviolet Interstellar Extinction

Proposal 10960: Moons of Mystery: Decoding the Aromatic and Aliphatic Organic Compositions of Phoebe, Hyperion, and Iapetus

Co- Principal Investigator:

Proposal 10308: Red Stars over PAHs: Hunting for PAH Emission in vdB 135, a Reflection Nebula Devoid of Ultraviolet Radiation

Skills:

Radiative Transfer (RADMC3D)

James Webb Space Telescope uncalibrated IFU NIRSpec & MIRI/MRS data reduction

Bayesian model fitting (PAHFIT (IDL and Python Version) & BAGPIPES)

MS Suite Proficiency

Excel, Python, & Shell Scripting (large database management and analysis)

Python & Fortran 90/95 Programming

Unix-based & Windows Operating Systems

Pyplot & Gnuplot Software (for model plotting)

Neural Networks (particularly for problem solving purposes)

Partial/Ordinary Differential Equations & Linear Algebra (including practical applications)

Numerical Methods & Computational Techniques

Mathematical Modeling of Physical Systems

Use of Analytical Problem-Solving Skills & Statistical Analysis

Component level computer construction, repair, update, and troubleshooting

Professional Societies:

Physics and Astronomy Graduate Student Association 2019-2025

Vice-President: 2021-2023

Sigma Pi Sigma National Physics Honor Society, 2018-Present

Society for Physics Students, 2014-2019

Awards/Scholarships:

- M. Stewart Scholarship (2020)
- Newell S. Gingrich Physics & Astronomy Endowment (2021)
- Rose Marie Dishman Endowed Scholarship in Physics (2022)
- Eli Stuart Haynes & Nola Anderson Haynes Scholarship (2023)
- Harry E. Hammond Teaching Assistant Award (2022 / 2023)
- Carl & Brynn Anderson Travel Award (2023)
- NTHU Postdoctoral Research Fellow (2025-2026)

References:

Dr. Daniel Harsono, Assistant Professor [ddharsono@gapp.nthu.edu.tw]

Dr. Aigen Li, Curators' Distinguished Professor of Astrophysics [lia@missouri.edu]

Dr. Ingrid Mann, Professor of Space Physics [ingrid.b.mann@uit.no]